AP Computer Science A
Ticket Out the Door Solutions
Set 5: Methods of the Math Class

Name _____ Period _____

Skill 5.1 Exercise 1
Complete each task independently. Use Java statements and write exact output where requested.
(a) Assume x is a nonnegative double that has already been initialized. Declare a double variable y and store the square root of x in it. double y = Math.sqrt(x);
(b) Use Math.pow to declare and initialize a double variable result with the value of 3 raised to the fourth power. double result = Math.pow(3, 4); The value stored is 81.0.
(c) Explain why this statement does not compile. Correct it so the complete absolute value, including the fractional part, is stored in k. int k = Math.abs(-127.5); Math.abs(-127.5) returns a double, which cannot be assigned directly to an int. Correction: double k = Math.abs(-127.5); This preserves the complete value, 127.5.
(d) Write the exact output of each statement. Include .0 when Java prints it. (i) System.out.println(Math.min(100, 200)); Output: 100 (ii) System.out.println(Math.max(100.0, 200)); Output: 200.0 (iii) System.out.println(Math.max(Math.max(100, 200), 50)); Output: 200 (iv) System.out.println(Math.sqrt(81)); Output: 9.0 (v) System.out.println(Math.round(-157.2)); Output: -157 (vi) System.out.println(Math.ceil(-157.7)); Output: -157.0 (vii) System.out.println(Math.floor(-157.7)); Output: -158.0 (viii) System.out.println(Math.ceil(157)); Output: 157.0

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AP Computer Science A
Ticket Out the Door - Student Exemplar
Set 5: Methods of the Math Class

Name _____ Period _____

Skill 5.2 Exercise 1	
Use Math.random() to declare and initialize one variable per row with the stated type and range.	
double r: 0.0 <= r < 1.0 0 included; 1 excluded.	double r = Math.random();
int card: 1 through 52 Both endpoints included.	int card = (int)(Math.random() * 52) + 1;
int number: 100 through 199 100 included; 200 excluded.	int number = (int)(Math.random() * 100) + 100;
int offset: -100 through 0 Both endpoints included.	int offset = (int)(Math.random() * 101) - 100;

Explain why (int)Math.random() * 6 + 1 always produces 1. Write a corrected expression that can produce any integer from 1 through 6.

Casting happens first, so the value from Math.random() becomes 0. Then 0 * 6 + 1 is 1. Multiply before casting:
(int)(Math.random() * 6) + 1

Skill 5.3 Exercise 1
Translate each expression into Java. Declare a double variable with the indicated name; you do not need to calculate its decimal value.
(a) d1 = 3π sin(187°) + |cos(122°)| Use Math.PI, Math.sin, Math.cos, and Math.abs. Convert each angle with Math.toRadians because sin and cos expect radians. double d1 = 3 * Math.PI * Math.sin(Math.toRadians(187)) + Math.abs(Math.cos(Math.toRadians(122)));
(b) d2 = (14.72 raised to the power 3.801) + ln(72) Use Math.pow and Math.log. The notation ln means natural logarithm (base e). double d2 = Math.pow(14.72, 3.801) + Math.log(72);

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AP Computer Science A
Ticket Out the Door Solutions
Set 5: Methods of the Math Class

Name _____ Period _____

AP FRQ-Style Application: Dice Challenge (Required)

Complete all parts. This original, scaffolded question combines Skills 5.1 and 5.2. Write Java statements that could appear inside a method; a class or method header is not required.

A game rolls two six-sided dice. Each die can show any integer from 1 through 6, inclusive. The program calculates the distance between the two rolls and a score. Use the variables from earlier parts when completing later parts.

(a) Simulate the rolls.

Declare and initialize int variables die1 and die2. Use a separate call to Math.random() for each die. Every face from 1 through 6 must be possible, with approximately equal likelihood.

```
int die1 = (int)(Math.random() * 6) + 1;
int die2 = (int)(Math.random() * 6) + 1;
```

(b) Calculate the distance.

Declare and initialize an int variable difference using Math.abs so it stores the nonnegative difference between the rolls. For example, rolls of 2 and 5 produce a difference of 3.

```
int difference = Math.abs(die1 - die2);
```

(c) Calculate and display the score.

The score is the square root of the sum of the squares of the two rolls. Use Math.pow and Math.sqrt to declare and initialize a double variable score. Then print difference and score on separate lines.

```
double score = Math.sqrt(Math.pow(die1, 2) + Math.pow(die2, 2));
System.out.println(difference);
System.out.println(score);
```

(d) Check the calculation.

Suppose die1 is 3 and die2 is 4. Write the two lines printed by part (c). Explain why score must be stored as a double even when its value is a whole number.

```
1
5.0
```

The difference is |3 - 4| = 1, and the score is sqrt(9 + 16) = sqrt(25) = 5.0. Math.sqrt returns a double, so this expression produces 5.0. It cannot be assigned directly to an int without a cast, which could lose the fractional part for other rolls.

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